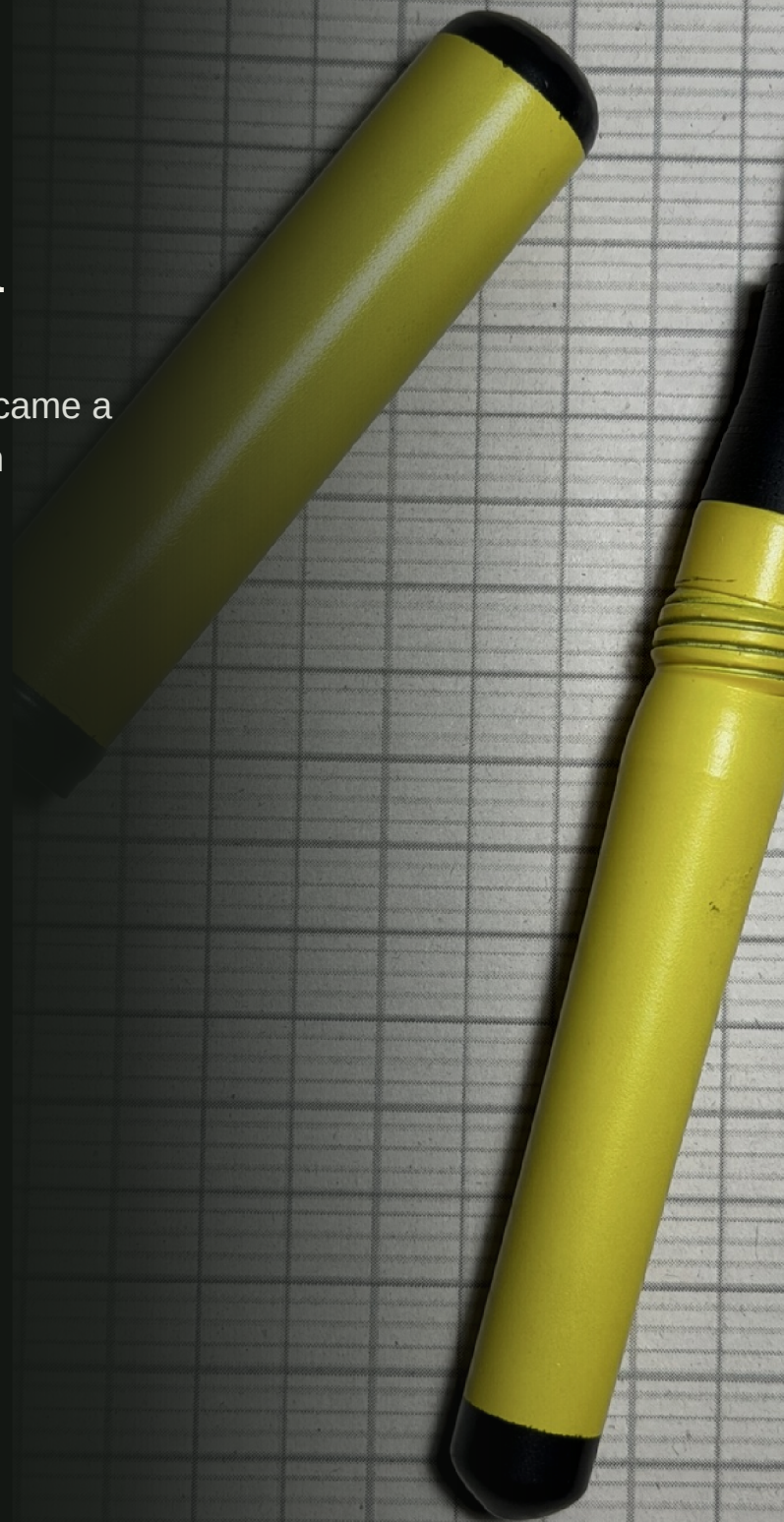


DESIGN RECORD | SEPTEMBER 24-OCTOBER 1,
2026

From first thread to finished pen

How a 149-inspired three-piece sketch became a print-tested, posting, modular fountain-pen platform.

Brian Johnson + Oracle
Fountain pen design history



A week of physical feedback shaped the design

The project began with a direct brief: create a three-piece, fully 3D-printed fountain pen with the visual presence of a classic 149, no clip or metal trim, a grip that accepts a Jinhao 100 #6 screw-in nib unit, and coarse threads that could survive repeated printing and use.

Measure -> model -> print -> test

Every important change came from either a real measurement, a printed-fit result, or a visible geometry problem. Digital checks narrowed the risk; Brian's printer and hand were the final authority.

1

CURRENT BARREL

6

COMPATIBLE CAPS

3

SECTION OPTIONS

The core interfaces stabilized first. J1 holds the nib unit inside the section. J2 joins section to barrel. The original J3 cap-to-section joint was retired when the architecture moved to J4, a cap-to-barrel joint. From there, the project could expand cosmetically without disturbing the proven mechanical core.

The result was not one uninterrupted CAD pass. It was a chain of small, testable corrections: compensate for FDM shrinkage; enlarge the smooth nib bore; move the nib thread behind the feed's required seating length; coarsen thread teeth that would not reproduce; check cap depth; remove a mouth interference; revise resin clearances; and reshape the barrel until it posted securely and looked continuous.



Physical generations exposed what the model alone could not: print resolution, joint feel, proportions, and assembly behavior.

V1 solved the interfaces one failure at a time

Sep 24 ● Start with a three-piece 149-inspired architecture

The first design separated section, barrel, and cap. The cap initially threaded onto the section through J3; the barrel joined the section through J2; the Jinhao unit engaged the internal J1 thread.

v1-v4 ● Use the nib thread as the first go/no-go gate

The first coupon printed about 0.4 mm undersize, so the FDM version gained +0.2 mm radial compensation. Brian's Jinhao unit then threaded into the coupon: the first proven interface.

v5-v6 ● Make the printed joints visible and usable

J2 and J3 received the same FDM compensation. J2 changed from fine 10.5 x 1.0 x 0.55 mm geometry to a chunkier 11.0 mm diameter, 1.5 mm pitch, 0.75 mm depth profile after the fine teeth failed to reproduce clearly.

v7 ● Move J1 behind the feed's smooth seating zone

Brian caught a functional conflict: the feed needed 13 mm of smooth bore from the nose. J1 moved from z=8-13.8 to z=13-18.8 so the housing could enter before its thread engaged.

v8-v9 ● Smooth the silhouette and soften the thread flanks

The abrupt section-to-barrel shoulder became a concave fillet and the barrel mouth gained a chamfer. Thread flanks softened from roughly 81 degrees toward a more forgiving 76-degree trapezoidal form, while flat crests, axial clearance, and coarse pitches were retained.

v10-v11 ● Size the bore from the real nib housing

A photograph and caliper reading established an 8 mm smooth housing. The bore grew first to 8.3 mm, then to an 8.7 mm design diameter so FDM shrinkage would leave about 8.3 mm in the print. The nose exterior grew to preserve wall thickness.

v12-v14 ● Resolve length, balance, and cap clearance

The barrel shortened from 100 to 97 mm and received a rounder tail. The cap lengthened from 51.5 to 62 mm after the nib extended 4 mm beyond the old cavity; the new interior left about 7.5 mm clearance. A smooth elliptical dome replaced the flat top.



Measurement translated the real nib unit into a printable section rather than relying on nominal dimensions.

The model followed the hardware, not the other way around

The Jinhao assembly set the internal constraints: an 8 mm housing, a 13 mm smooth insertion path, and a threaded collar that had to start only after the feed seated. That forced a larger bore and a thicker nose wall.

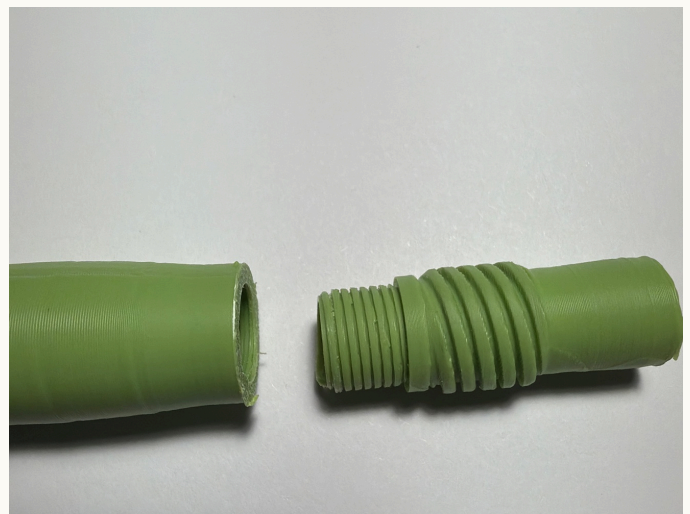
The joints were treated differently by purpose. J1 remained fine enough to match the manufactured nib unit. J2 became a single-start, coarse 1.5 mm-pitch connection. The cap joint used multiple starts for quicker closure.

The first major milestone was simple and decisive: “It worked.”

Once the J1 coupon passed, the project could build around a proven core. Brian later reported that the complete printed parts fit “really well,” confirming the section, barrel, and cap as an assembly.

WHAT THE EARLY PRINTS TAUGHT US

- Nominal CAD clearance is not printed clearance.
- Fine threads can disappear even when their geometry is mathematically valid.
- Axial access matters: a thread can fit at its destination yet block the part before it gets there.
- Wall thickness must be rechecked whenever an interior bore grows.



Coarser teeth and additional clearance made the printed interfaces legible and serviceable.

V2 moved the cap to the barrel - then exposed a hidden jam

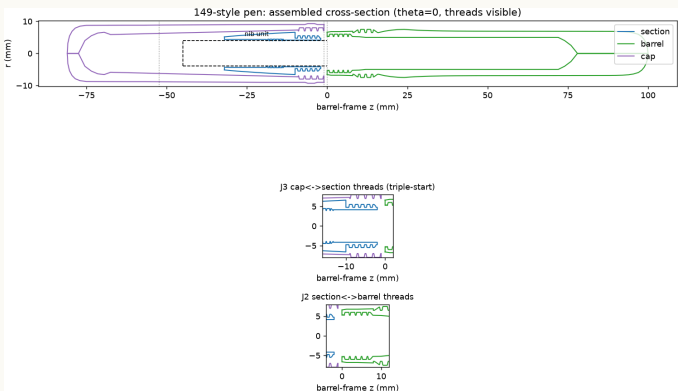
The traditional redesign shortened the section from 40 to 30 mm, removed J3 completely, and moved J2 to the rear of the shorter section. A smooth shoulder flared to meet the barrel mouth. The cap now screwed to the barrel through J4, a 15 mm, 2.0 mm-pitch, triple-start thread.

The first V2 cap looked correct in its seated position but could not actually pass over the barrel's 15.0 mm male teeth: its mouth bore was only 14.6 mm. Moving the female thread to the mouth fixed the obvious obstruction, but a tapered runout still left an undeveloped entrance that jammed during the screw-on path.

Static fit was necessary, not sufficient

The seated surfaces showed 0.450 mm radial clearance. Only an insertion simulation revealed roughly 0.55 mm of interference at the entrance.

The final fix carried the female J4 profile fully to the cap mouth, using the male thread's own 2 mm runout as the lead-in. A permanent `check_thread_assembly()` gate was added so later joints would be tested across the real screw-on path, not only at their final position.



The section, barrel, and cap were checked as an assembly, with joint geometry inspected in context.

Joint	Role	Outcome
J1	Nib unit to section	Proven core; retained
J2	Section to barrel	Coarsened; later loosened
J3	Cap to section	Retired in V2
J4	Cap to barrel	Triple-start; full-profile mouth

Resin feedback turned a working pen into a balanced system

Sep 25 ● Separate FDM compensation from resin behavior

The first resin run showed J2 was far too tight. The FDM shrinkage allowance had made the male thread oversized on the more dimensionally faithful resin print. This corrected the assumption that the earlier clearance would simply feel looser in resin.

Sep 26 ● Simplify the barrel exterior

The mid-body bulge and hard step were removed in favor of a cleaner taper. An early threaded rear posting band was explored and then retired; Brian did not want another visible threaded feature at the tail.

Sep 26 ● Make posting a taper-wedge interaction

The replacement used the barrel body itself as a friction post. The barrel and cap-bore tapers were tuned so the cap seated at roughly three-quarters depth while its J4 mouth teeth remained clear.

Sep 26 ● Replace the kink with one continuous Bezier curve

Brian marked a visible transition at z=35 and asked for a single smooth curve. The final cubic Bezier runs from the J4 base toward a soft dome, changing from 15.0 mm diameter to 12.8 mm over a 93 mm barrel.

Sep 26 ● Retune caps and loosen production joints

The V3 and V2 cap bores were adjusted to seat at 76.9% and 77.4% of their available depth. J2 and J4 radial clearances increased from 0.45 to 0.50 mm for print variation; the proven J1 remained at 0.45 mm.

Brian's print verdict on the new set: “pretty amazing.”

93 mm

CURRENT BARREL

76.9%

V3 POST DEPTH

0.50 mm

J2 AND J4 RADIAL GAP

A stable core unlocked a family of interchangeable parts

Once the barrel, J1, J2, J4, cap bore, and posting behavior were stable, changes could be isolated to exterior form or refill support. By September 30 the platform comprised one barrel, six caps, and three sections.



Fluted cap. Long shallow channels, with the V3 mouth and interior retained.



Square cap. Beveled flats blended back to the round J4 mouth.



Pisa cap. Five raised floor rings and 32 shallow ribs abstract the tower.

SIX CAP OPTIONS

V3, V2, octagon, fluted, square, and Pisa. Each keeps the proven J4 interface; the exterior can change without rewriting the attachment system.

FLARED FOUNTAIN-PEN SECTION

A Leonardo-inspired trumpet flare widened the nib end to 11.9 mm, curved through a 10.81 mm waist, and met the existing shoulder without a kink. It retained the proven internals and printed support-free, screw-side-up.

BALLPOINT SECTION

A Parker G2/Quinkflow variant added a 2.75 mm pilot, an internal funnel and positive shoulder stop, plus a 10.08 mm press-fit rear plug. It used the same barrel and caps, though final tip protrusion still awaited the physical shoulder-to-ball measurement.

MODULARITY RULE

New variants had to preserve compatibility: cap clearance around the section, wall thickness, watertight meshes, joint fits, and posting behavior were all rechecked.

The toolchain connected calculation, visualization, and print reality

Measure

Generate

Validate

Preview

Print

1

make_pen.py - Python generator and validator

Using NumPy and Trimesh, the script computes the J1/J2/J3/J4 thread geometry, generates meshes, checks watertightness and clearances, and runs fit gates including the real insertion-path simulation in **check_thread_assembly()**

. It outputs the ASCII STL parts used for delivery.

2

pen149.scad - OpenSCAD parametric master

Brian selected OpenSCAD as the going-forward design tool. The master exposes parts and modes parametrically, and supports visual verification of thread fits and assembly, including importing exported STLs back into an assembly view.

3

ASCII STL - reliable delivery format

Binary STL and 3MF files arrived corrupted on Brian's Mac, sometimes reporting no geometry. ASCII STL became the dependable handoff format. Resolution settled at 224 facets because a 448-facet build was too large and finer than a 0.4 mm nozzle could resolve.

4

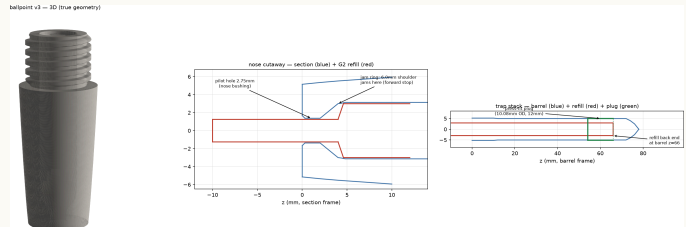
Matplotlib true-geometry previews

Small render scripts - `render_cap.py`, `render_cap_pisa.py`, `render_cap_square.py`, and `render_section_ballpoint.py` - visualize the actual STL geometry and cutaway relationships. They are inspection views, not substitutes for a print.

5

AI image generation for finish decisions

Generated colorway concepts and decoration mockups helped compare ideas before resin, paint, or UV ink was committed. Examples include the yellow-and-black scheme and bee-wrap previews applied to Brian's finished-pen photo. These guided appearance only; the engineering geometry remained deterministic.



A typical engineering preview combines a mesh view with section and assembly relationships.

Physical use closed the loop that validation began



The finished V3: black resin, copper/bronze fleck, and eufyMake E1-printed Muse mascot graphics with a white underbase.

On September 28 Brian completed the V3 in black resin with copper/bronze flecks and printed graphics on the cap. After living with the finished object, his assessment covered the two behaviors the design had worked hardest to secure:

“Posts beautifully and it's a dream to write with.”

That result validated more than dimensional fit. The cap stayed on the tail at the intended depth, the pen balanced in use, the section accepted the nib unit, and the finished surface supported decoration.

A second finish study moved from generated color concepts to a real yellow pen. The chosen scheme kept the body yellow, made the first 7 mm of the cap top and barrel bottom black, added a thin black line at the cap base, and used a black grip. Brian's October 1 photograph matched the preview closely.

Preview before committing material

The later workflow treated finish mockups like fit coupons: a low-cost way to test a decision before using resin, paint, or eufyMake ink.

The current platform is both finished and extensible

The V3 cycle reached a complete object: it fits, posts, writes, and accepts decorative finishes. At the same time, the mechanical interfaces remain a platform for future cap, section, and refill experiments.

1

BEZIER WEDGE-
POST BARREL

6

CAP GEOMETRIES

3

SECTION
GEOMETRIES

DURABLE DESIGN LESSONS

- 1. Measure the real component.** Nominal nib dimensions were not enough to establish the housing bore and feed path.
- 2. Validate the motion, not just the final pose.** The V2 cap could pass a seated clearance test and still jam on the way in.
- 3. Tune for the printing process.** FDM compensation improved one print and made a resin joint too tight.
- 4. Lock interfaces before multiplying styles.** Stable J1, J2, J4, and posting geometry made six caps and three sections manageable.
- 5. Use the print as evidence.** Fit, feel, posting depth, and writing quality remain the final validation.

Status at October 1, 2026. The ballpoint variant remains subject to Brian's final refill shoulder-to-ball measurement; the fountain-pen platform itself has completed a successful design-to-finish cycle.



The latest finished build shows the mature geometry carrying a deliberately previewed color scheme.